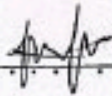


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COMP 009 Object-Oriented Programming

Activity #1

(See photo below)



Activity #1 (OOP)

PART A - VARIABLE IDENTIFICATION TABLE

VARIABLE NAME	DATA TYPE	VARIABLE TYPE	STORED IN
name	String	Instance	Heap
grade	int	Instance	Heap
n	String	Parameter	Stack
g	int	Parameter	Stack
args	String	Parameter	Stack
s1	Object	Reference	Stack
s2	Object	Reference	Stack
s3	Object	Reference	Stack

PART B - EXPLANATION

STATEMENT	WHAT HAPPENS?	MEMORY IMPACT
student s1 = new student ("John", 85);	Creates new student object with name "John", and grade of 85. Reference being s1.	Object is created in heap. Reference is stored in stack.
Student s2 = new students ("Maria", 90);	creates another object with name "Maria" and a grade of 90. Reference s2.	New object in heap. Reference in stack.
student s3 = s2;	Copies the reference of s2 storing into s3.	s3 is stored in stack pointing to heap

STATEMENT	WHAT HAPPENS	MEMORY IMPACT
s1.grade = 95;	Updates the grade of object referenced by s1.	changes existing heap. No new memory added.
s3.name = "Ana";	changes name of the object referenced by s3, which changes s2 as well.	modifies Heap shared by s2 and s3.

PART C- OUTPUT

Source Code:

John 95

Ana 90

Ana 90